Telomy - Detailed Project Report (DPR)

Aging-Intelligence Models, SLM & Platform

Version: 2.0 (FINAL) * **Date:** 2026-10-21 * **Owner:** ananshubapu@gmail.com **Scope:** every dataset, formula, model, workflow, agent, architecture, roadmap, open item and business opportunity behind Telomy's aging-intelligence stack. All metrics are measured and reproducible, not projected. Sec.9 lists what is done vs pending.

---

0. Executive summary

Telomy estimates a person's **biological aging** from an inexpensive routine blood panel (and, increasingly, from wearables), then explains it through an **on-device, tool-grounded small language model (Telomy-LLX)** that never invents a number. The clocks are trained on **NHANES 2009-2018** (49,693 participants). The flagship clock predicts chronological age at **MAE 8.8 years (r=0.80)** - at parity with the best published routine-bloodwork clocks - and the mortality-validated **PhenoAge** biological-age measure is reproduced at **r=0.92**. The moat is not raw accuracy; it is **accuracy-per-rupee + multimodal fusion (bloods x wearables) + on-device privacy + a shipped clinical operating system.**

---

1. Datasets

1.1 NHANES (primary training corpus)

- **Source:** US CDC National Health and Nutrition Examination Survey, public-domain.
- **Cycles used:** 2009-10 (F), 2011-12 (G), 2013-14 (H), 2015-16 (I), 2017-18 (J) - 5 continuous cycles.
- **Files:** 46 SAS-XPT lab/demographic files across 11 families: `DEMO` (age, sex, race), `CBC` (haematology), `BIOPRO` (comprehensive metabolic/biochemistry), `GLU` (fasting glucose), `GHB` (HbA1c), `HSCRP`/`CRP` (inflammation), `INS` (insulin), `TCHOL`, `HDL`, `TRIGLY` (lipids).
- **Assembled table:** 49,693 participants x 51 columns, keyed by `SEQN`, cycles concatenated.
- **Modelling cohort:** adults 18-80 (NHANES top-codes age at 80); **~27,000 with full blood panels**, 16,396 with CRP. This is **27x the 1,000-samples-per-model requirement.**
- **Download note (ops):** CDC blocks non-browser user-agents; fetch with a browser UA from `wwwn.cdc.gov/Nchs/Data/Nhanes/Public/<startyear>/DataFiles/<FILE>_<suffix>.xpt`.
- **Harmonisation:** CRP unified to mg/L (older cycles report conventional CRP in mg/dL x10 -> hs-CRP mg/L); glucose unified (fasting where available, else serum).

1.2 NHANES body composition / muscle (muscular-age clock)

- **Files:** DXA whole-body (`DXX`, 2011-2018), grip strength (`MGX`, 2011-2014 only - discontinued after), body measures (`BMX`, all cycles), demographics (`DEMO`). 23,825 adults assembled; DXA on 22,511; grip on 10,386.
- **Derived clinical features:** appendicular lean mass ALM = sum(arm+leg lean, excl. bone); **ALMI = ALM / height2** (the sarcopenia index); total/trunk lean, fat %, lean:fat ratio, bone mineral density, grip strength, anthropometry.

1.3 Nutrition - USDA FoodData Central (meal-photo feature)

- **SR Legacy** bulk CSV -> compact per-food table of 7,793 foods with macros + micronutrients per 100 g (`telomy-vision/assets/nutrition_db.csv`). Grounds meal-photo nutrition the same way the clocks ground age (model proposes the food, the table supplies the numbers).

1.4 NHANES diagnoses + linked mortality (disease & long-term models)

- **Diagnosis questionnaires** (2009-2018): diabetes (`DIQ`), BP/cholesterol (`BPQ`), medical conditions (`MCQ`: CHD, MI, angina, CHF, stroke, liver, cancer, COPD), kidney (`KIQ_U`). 30,352 adults joined to the blood panel by SEQN -> 12 physician-diagnosed binary labels.
- **NHANES linked-mortality public files** (follow-up through 2019, fixed-width): SEQN, MORTSTAT, UCOD_LEADING, PERMTH - enables long-term outcome modelling. 2009-2012 cycles carry the 8-10y follow-up needed for a clean 10-year label.

1.5 Telomy device data (operational, feeds future models)

- **KY-E-B1 ECG patch** - per-minute HR/HRV (RMSSD, SDNN), beat counts, quality, arrhythmia events; live to cloud (`band_minutes`, `band_events`, `ecg_strips`).
- **Smart ring (TK-series)** - PPG HR, SpO2, motion, overnight sleep (`vitals_ts source=ring`).
- **WHOOP 4.0** - HR/SpO2/respiration/skin-temp 1 Hz overnight, sleep staging (via NOOP export).
- **System of record:** Oracle cloud SQLite (`/opt/telomy/telomy.db`), nightly backups.
- These are **longitudinal per-person trajectories** - the substrate for the future pace-of-aging model (NHANES cannot provide this, being cross-sectional).

1.3 Reference/validation assets

Published implementations used for cross-checking: `BioAge` (R; NHANES KDM/PhenoAge/homeostatic dysregulation), `pyaging` (pretrained KDM). UK Biobank literature informs feature choice (cystatin C).

---

2. Formulas & rationale

2.1 Levine PhenoAge (biological age, mortality-validated) - reproduced exactly

Inputs (9 markers + chronological age), converted to the formula's units: albumin->g/L (x10), creatinine->umol/L (x88.4), glucose->mmol/L (x0.0555), ln(CRP in mg/dL), lymphocyte %, MCV (fL), RDW (%), ALP (U/L), WBC (103/uL), age (yr).

Linear predictor:

```
xb = -19.907
     - 0.0336*albumin + 0.0095*creatinine + 0.1953*glucose + 0.0954*ln(CRP)
     - 0.0120*lymph% + 0.0268*MCV + 0.3306*RDW + 0.00188*ALP + 0.0554*WBC + 0.0804*age
mortality = 1 - exp( -exp(xb) * (exp(0.0076927*120) - 1) / 0.0076927 )
PhenoAge  = 141.50225 + ln( -0.00553 * ln(1 - mortality) ) / 0.090165
AgeAccel  = PhenoAge - chronological_age        <- the clinical signal (+/-0 = on pace)
```

Rationale: PhenoAge is a Gompertz/Cox mortality model re-expressed in age units; it predicts all-cause and cause-specific mortality *beyond* calendar age. Telomy reports **AgeAccel** - the number that actually responds to disease and lifestyle. Reproduced at **r=0.917** vs chronological age (Levine reported r~0.9).

2.2 Telomy learned clocks (chronological-age regression)

- **Algorithm:** `HistGradientBoostingRegressor` (sklearn) - gradient-boosted histograms, **native missing-value handling** (critical: real panels are incomplete; no imputation bias), L2 regularisation. Chosen over XGBoost/LightGBM for equivalent accuracy without the `libomp` native dependency, and over linear models because ElasticNet baselines run 2.5-3 years worse on MAE.
- **Target:** chronological age (NHANES `RIDAGEYR`), adults 18-80.
- **Validation:** 20% held-out test + 5-fold cross-validation; metrics MAE (years), R2, Pearson r.
- **Rationale for a family of clocks** (not one): a full panel is most accurate, but organ-/panel- specific sub-clocks let us (a) work from whatever a clinic measured, (b) localise aging to a system (metabolic, hepatic/renal, haematologic), which is clinically actionable and explainable.

2.3 Explanation layer

Each measured marker is compared to an adult reference median; deviation >12% is reported with its known direction of age-association (e.g. high glucose/RDW/CRP -> older; high albumin/lymph%/HDL -> younger). Transparent and assay-only - no black-box attribution required.

---

3. Models - how each works (detail)

Model	Inputs	N	Test MAE	R2	Pearson r	Role
telomy_full	37 routine markers + sex	27,340	**8.79 y**	0.639	0.800	flagship chronological clock
telomy_metabolic	glucose, HbA1c, insulin, lipids, urate, BUN	27,327	11.12 y	0.434	0.659	metabolic sub-clock
telomy_liverkidney	LFTs + renal markers	27,280	12.08 y	0.358	0.598	hepatic/renal sub-clock
telomy_clinic9	PhenoAge's 9 markers	27,791	12.20 y	0.357	0.597	cheap-panel clock
telomy_cbc	13 haematology markers	27,733	13.31 y	0.241	0.491	haematologic sub-clock
PhenoAge (formula)	9 markers + age	16,396	-	-	r=0.917 vs age	biological age + AgeAccel

Top age-driving markers (|corr| with age): glucose (+0.22), albumin (-0.21), MCV (+0.21), creatinine (+0.18), RDW (+0.17), lymphocyte% (-0.15), ALP (+0.12).

Muscular-age clock family (NHANES DXA + grip + anthropometry, 2011-2018)

Same method (HistGradientBoosting, 20% holdout + 5-fold CV), target = chronological age; residual (predicted - actual) = **muscular-age acceleration** (the sarcopenia signal).

Model	Inputs	N	Test MAE	R2	Pearson r
muscle_full	DXA + grip + anthropometry (19 features)	21,610	**9.05 y**	0.606	0.779
muscle_dxa	DXA + anthropometry (no dynamometer)	21,668	9.51 y	0.553	0.745
muscle_anthro	scale + tape only (phone-feasible)	22,533	11.93 y	0.385	0.620

Dominant feature: **ALMI (appendicular lean mass index), r = -0.62 with age** - the clinical sarcopenia marker; grip strength r = -0.32. Code: `telomy-app/research/muscleclock/`.

Meal-photo nutrition (telomy-vision)

`image -> food_recognition (CLIP zero-shot label) -> portion estimate -> nutrition.lookup()` grounded in USDA SR Legacy (7,793 foods). Returns per-serving + meal-total macros/micros. Same provenance rule as the clocks: the vision model proposes, the USDA table supplies the numbers.

Medical-image / report ingestion (telomy-vision/imaging) - decision-support only

DICOM read (pydicom) + report OCR/structuring (pdfplumber/pytesseract); pixel-level assist via **pretrained** published models (torchxrayvision chest-X-ray, TotalSegmentator/MONAI for CT/MRI), clinician-in-the-loop. No autonomous diagnosis, no from-scratch diagnostic model. Genome = parse the lab report/VCF (sequencing is not images).

Disease-prediction models (NHANES diagnoses, routine panel + age/sex)

Per-condition HistGradientBoosting classifiers; 20% held-out test ~ **6,000 reports each**; metric ROC-AUC. Match or beat the published NHANES literature.

Condition	AUC	Condition	AUC	Condition	AUC
diabetes*	0.887	heart attack	0.842	hypertension	0.823
heart failure	0.858	COPD	0.826	kidney disease	0.803
coronary heart disease	0.849	cancer	0.811	angina	0.793
stroke	0.792	high cholesterol	0.770	liver disease	0.731

* diabetes scored **without** glucose/HbA1c (non-circular); with them AUC 0.929. A "guess the disease" ranker scores all conditions for one report (ranked coronary HD #1 at 0.85 for a known-CHD patient).

Long-term models - mortality (NHANES linked-mortality, follow-up to 2019)

- **10-year all-cause mortality: test AUC 0.885** (NHANES 2009-2010 only - the single cycle with a full 10-year window, so no follow-up-length label leakage; n=3,607, 766 deaths, test 722).
- **Cardiovascular-cause death: test AUC 0.868** (heart + stroke; 583 events, all cycles pooled). Follow-up varies 1-10y by cycle, so a calibrated time-to-event (Cox / competing-risks) model is the rigorous upgrade - labelled as such.
  Age-dominated; needs calibration + benchmarking vs Pooled Cohort Equations before clinical use.

Rigor pass (disease models) - `diseasemodel/eval_rigor.py`

Calibration + PR-AUC + a multi-label "guess the disease" metric. Finding: **ranking is strong** (guess-the-disease **top-3 recall 92.3%**, top-1 70.9%) but **absolute probabilities for rare conditions are miscalibrated** (balanced class weights trade calibration for ranking) -> present as ranked risk and add `CalibratedClassifierCV` (isotonic) before showing percentages. High-prevalence conditions (diabetes/hypertension/cholesterol) are well-calibrated (calib-err 0.10-0.17). Code: `telomy-app/research/diseasemodel/`.

Unified report flow - one panel -> the whole picture

`telomy-slm/telomy_llx/report.py::analyze_report(panel)` orchestrates every grounded tool into one narrative: **blood age (clock) + biological age/acceleration (PhenoAge) + muscular age (if body composition present) + ranked disease screen + 10-year mortality risk**. Verified end-to-end; e.g. a 58-yo panel -> blood age 61, biologically +13y, muscle -10y, flags hypertension/stroke/liver, 10y risk 4%.

Telomy-LLX (the SLM) - tool-grounded longevity model

- **Base:** Qwen2.5-3B-Instruct (cloud) / Qwen2.5-1.5B-Instruct (on-device, 4-bit MLX/llama.cpp).
- **Grounded tools:** aging (`predict_age`, `phenoage`), disease (`predict_diseases`, `mortality_risk_10y`), muscle, nutrition - the model calls them and explains; never computes.
- **Core principle:** the model **never computes a number**. It emits function calls (`predict_age`, `phenoage`, `explain_drivers`); the tool layer runs the *trained clocks + PhenoAge*; the model then explains the returned numbers with provenance. This structurally eliminates the #1 failure mode of medical LLMs (fabricated figures).
- **Training:** QLoRA SFT on grounded conversations auto-generated from NHANES (user panel -> tool calls -> tool results -> faithful explanation), measured for **numeric-faithfulness** (every number traces to a tool) and **guardrail-rate** (defers to clinicians, no diagnosis/prescription).
- **Privacy:** runs on-device; patient data never leaves the phone (DPDP/GDPR USP).
- **Status (measured):** foundation shipped in `telomy-slm/` - grounded tool layer runs; 2,700/300 SFT conversations generated from NHANES; offline eval **100% guardrail-rate, 93.3% numeric- faithfulness** (dataset grounded by construction; strict checker flags small integers in prose). QLoRA training entrypoint + configs ready for a GPU box; fine-tune + on-device export pending.

Telomy Health Agent - fusion engine + safety workflow (telomy-slm/telomy_llx)

One orchestrator (`agent.py::TelomyHealthAgent`), two jobs, 9/9 tests passing (`tests/test_agent.py`): - **Wearable-fusion risk engine** (`fusion.py`): fuses the trained blood models with a **cited wearable overlay** (resting HR, HRV/SDNN, nocturnal SpO2 - published hazard ratios, Zhang 2016 / Dekker 2000; labelled an evidence overlay, NOT a trained model, upgradeable to a trained fusion head on device-cohort outcomes). Surfaces blood-biological-age AND wearable convergence - a multimodal signal rivals lack. - **Safety workflow** (`falldetect.py` + `monitor.py`): 3-phase fall detection (impact **and** post-impact immobility - rejects "sat down hard"; real fall 0.98 conf) feeding an emergency state machine (NORMAL->CONFIRMING->ESCALATED) with a confirmation countdown, immobility gate and user override. Escalation **prepares** an alert payload; the app sends it to emergency contacts with consent - the agent never messages anyone. Triggers: high-confidence fall, SpO2<88, HR>150 at rest or <35. Full design: `telomy-slm/AGENT.md`.

4. Workflows & architecture

4.1 Model training pipeline (`telomy-app/research/ageclock/`)

```
CDC NHANES XPT  ->  assemble.py  ->  nhanes_bloodpanel.csv  ->  train.py  ->  models/*.joblib + results.json
(46 files, 5 cycles)   merge on SEQN,        49,693 x 51         HGB + 5-fold CV,     5 clocks + metrics
                       rename, harmonise                          PhenoAge reimpl.
```

4.2 SLM pipeline (`telomy-slm/`)

```
models/*.joblib -+
                 +- tools.py (grounded tool layer: predict_age / phenoage / explain_drivers + schemas)
NHANES csv ------+        |
                          +- datagen.py -> data/sft_{train,val}.jsonl (grounded tool-call conversations)
                          +- train_qlora.py (Qwen2.5 QLoRA, GPU) -> LoRA adapter
                          +- eval.py (numeric-faithfulness + guardrail-rate)
                          +- export -> 4-bit GGUF/MLX -> ships inside the iOS app beside tools.py + clocks
```

4.3 Product/platform architecture (already shipped & verified)

```
iOS/RN app --(HTTPS)-- FastAPI backend (Oracle cloud) -- SQLite (per-user) + care graph + billing
   |  ring/patch/WHOOP -> vitals, HRV, sleep, ECG, aging clocks, 3D twin
   +- roles: user * doctor * centre * super-admin   (tenant-isolated, token auth, 401/403 enforced)
```

The clocks and Telomy-LLX plug into this: the app calls the clock endpoints; Telomy-LLX runs on-device and explains results; the web console surfaces aging dashboards to clinicians.

4.4 Autonomous intelligence agents (continuous)

Two scheduled agents run every ~4 hours whenever the Mac is on and online (catch up on next launch), appending to `telomy-app/research/intel/`: - **`telomy-research-intel`** - rotating 15-theme deep-research sweep (datasets, model SOTA, exposures, cardiac/fall/long-term models, regulatory, genomics, funding, IP) -> `FINDINGS.md`, each entry ending in "Standout opportunities for Telomy". - **`telomy-market-intel`** - studies one rival platform per run -> updates `FEATURE_MATRIX.md` (basic/advanced features x who-has-it x Telomy status) and appends profiles to `MARKET_FINDINGS.md` with "What Telomy should integrate". Both do web research only - no external sends, no spend. Review cadence: every 2 days. Disable in the app's Scheduled section when the sweeps are complete.

5. Future development plan

Horizon	Work	Target / rationale
0-90 days	Add **cystatin C** to the panel & retrain	UK Biobank's single most predictive marker (PhenoAge omits it); push MAE <8 y
0-90 days	**Wearable-fusion features** (resting HR, HRV, arrhythmia burden) into the clocks	a cohort hit 5.68 y MAE by adding biometrics; we already collect these live
0-90 days	**Telomy-LLX v0** fine-tune (Qwen2.5-3B) + on-device 1.5B build	tool-grounded longevity copilot in-app
3-6 mo	**External validation** on UK Biobank / Indian cohorts; recalibrate for Indian reference ranges	generalisation + local regulatory evidence
3-6 mo	**Mortality/disease-risk heads** (Cox) on top of the panel	move from "age" to "risk" - higher clinical value
6-12 mo	**Pace-of-aging model** - latent Neural-ODE / Universal Differential Equation on Telomy's own longitudinal device+biomarker trajectories	the honest "PINN" - needs per-person time-series we're now accruing
6-12 mo	**Clinical study / CDSCO pathway**; prospective accuracy & utility	credibility + reimbursability
12 mo+	**Multi-omics option** (methylation add-on) for premium tier	<3 y MAE where users pay for it

Honest non-claims (diligence hygiene): we do not beat methylation clocks on raw MAE today; the pace-of-aging ODE is gated on longitudinal data accrual; Telomy-LLX is scoped and wired, not yet fine-tuned.

---

6. Business opportunities

1. **B2C longevity subscription** - "know and lower your biological age" from a cheap blood panel + wearable; recurring revenue; the on-device copilot is the retention engine.
2. **B2B2C via clinics & wellness centres** - the shipped doctor/centre/super-admin platform is the distribution channel; aging dashboards + AgeAccel tracking as a paid clinician tool.
3. **Corporate wellness** - employer programs priced on measurable biological-age improvement (clear ROI story vs step-count wellness).
4. **Insurance / risk underwriting** - PhenoAge AgeAccel and the risk heads are directly relevant to life/health underwriting and wellness-linked premiums (partnership or data product).
5. **Pharma & longevity-trial endpoints** - biological-age change as a surrogate endpoint; CRO/biotech licensing of the validated clock + fusion data.
6. **API / white-label** - "biological age as an API" for labs, telehealth, and other apps; the tool layer is already an API-shaped boundary.
7. **Diagnostics tie-in** - bundle with partner lab panels (incl. cystatin C) for an integrated "Telomy Longevity Panel" product.
8. **Data network effect** - the multimodal longitudinal dataset (bloods x wearables x outcomes) is itself a compounding, defensible asset few can replicate at consumer scale.

Primary wedge: B2C app + clinic distribution now; insurance/pharma data partnerships as the dataset and validation mature.

---

7. Status - done vs pending (as of 2026-10-21)

Done & verified (reproducible, tested)

- Blood aging-clock family (5 models; flagship MAE 8.79y / r 0.80) + PhenoAge (r 0.917).
- Muscular-age clock family (MAE 9.05y; ALMI -0.62 with age).
- Disease-prediction models - 12 conditions (AUC 0.73-0.89), tested on ~6k reports each; "guess the disease" ranker.
- Long-term mortality: 10-year all-cause (AUC 0.885) + cardiovascular-cause death (AUC 0.868, 583 events).
- Nutrition engine - USDA SR Legacy + IFCT 2017 India (8,335 foods) + 24-dish Indian recipe resolver.
- Vision - meal-photo -> nutrition and chest-X-ray decision-support proven end-to-end on CPU.
- Genome annotation with real gnomAD South-Asian allele frequencies.
- Telomy-LLX foundation - grounded tool layer, 2,700/300 SFT conversations, eval 100% guardrail / 93.3% faithfulness.
- Unified report flow + Telomy Health Agent (fusion engine + fall/emergency safety), 9/9 tests.
- Two continuous intelligence agents + the deep-tech org (handbook, 7-agent team, enabled plugins).

Pending (gated on GPU, data, ops, or partnerships - not blocked on core research)

1. **Telomy-LLX fine-tune + on-device export** - GPU-gated; data/entrypoint/configs ready.
2. **Wearable-fusion -> trained head** - needs the Telomy device cohort to accrue outcomes; today it's a cited evidence overlay.
3. **Fall/emergency -> native + validation** - port `detect_fall` to Swift/CoreMotion; wire consent-based contact delivery in the app; validate on real-fall data (FARSEEING), report false-alarms/day.
4. **Clinical & regulatory** - add cystatin C -> retrain; external Indian-cohort validation with calibration vs Pooled Cohort Equations; CDSCO / FDA-SaMD pathway; DPDP/GDPR file.
5. **Pace-of-aging ODE** - gated on longitudinal per-person device data accrual.
6. **Operational parity (from competitive intel)** - lab home-collection, e-pharmacy, Rx pathways, CGM, imaging: partnerships, not R&D.
7. **Recipe DB breadth & food classifier** - expand beyond 24 dishes; train a cuisine classifier (GPU).
8. **Change cloud super-admin password**; decide cloud-lock vs Mac for device studies.

Known limitation documented

- One pre-existing therapy-verdicts test is flaky under the shared test DB (passes in isolation); unrelated to the model work.

8. Reproducibility index

- Blood clocks: `telomy-app/research/ageclock/{assemble.py, train.py, results.json, models/}`; data `raw/`.
- Muscular clock: `telomy-app/research/muscleclock/{muscle_clock.py, results.json, models/}`.
- Disease + long-term: `telomy-app/research/diseasemodel/{assemble_disease.py, train_disease.py, mortality_longterm.py, eval_rigor.py, results*.json, models/, README.md}`.
- Strategy docs: `telomy-app/research/{LONGEVITY_INTERVENTIONS.md, CLINICAL_REGULATORY.md}`; intel in `research/intel/`.
- SLM + unified flow + health agent: `telomy-slm/{telomy_llx/ (tools, disease_tools, report, fusion, falldetect, monitor, agent, datagen, train_qlora, eval), configs/, data/, tests/, README.md, AGENT.md}`.
- Vision/nutrition/imaging: `telomy-vision/{telomy_vision/, imaging/, assets/nutrition_db.csv, assets/ifct2017/, README.md}`.
- Intelligence agents: `telomy-app/research/intel/{AGENDA, FINDINGS, COMPETITORS, FEATURE_MATRIX, MARKET_FINDINGS}.md`; scheduled tasks `telomy-research-intel` + `telomy-market-intel`.
- Org & team: `telomy-app/{ENGINEERING.md, CLAUDE.md, .claude/agents/}`.
- Companion: `telomy-app/research/ageclock/REPORT.md` (research landscape + investor framing).